23CP309P- Big Data Analytics Lab

End Semester Practical Examination, Date: 29/04/24 (SET B)

Implementation time is 2 hours

Question- 2. The file has the following features "rate_marriage", "age", "yrs_married", "children", and "religious". The output feature, "affairs", has binary values (0 for no affair, 1 for affair). The sample data is shown below.

rate_marriage	age	yrs_married	children	Religious	affairs
5	32	5	2	2	1
4	31	4	1	1	1
3	29	4	0	3	1
3	28	2	2	1	0
4	31	3	1	3	1
2	25	1	0	2	1
3	27	2	1	3	0
5	30	7	2	2	0
4	23	2	0	1	1

Perform the following operation in PySpark

- Create a dataframe for the above sample (3 marks)
- Group by rate_marriage, count and display output (2 marks)
- Group by affairs, count and display output (2 marks)
- Group by rate_marriage, affairs and perform count, perform order by rate-marriage, affairs, count in ascending order and display the output. (4 marks)

- Create the vector assembler and transform the dataframe (5 marks)

- Feature Selection (6 marks):
  - Use feature selection techniques (such as correlation analysis or feature importance) to select the most relevant features for classification
- Model Building (8 marks):
  - Split the dataset into training and testing sets.
  - Build and train a machine learning classifier (such as logistic regression, decision tree, or random forest) to classify ~~iris flowers~~ affairs into their respective ~~species.~~ data
  - Evaluate the performance of the model using appropriate evaluation metrics (such as accuracy, precision, recall, or F1-score).